Lecture 22

Equilibrium figures

From last time…

Lagrangian in a self-gravitating planet:

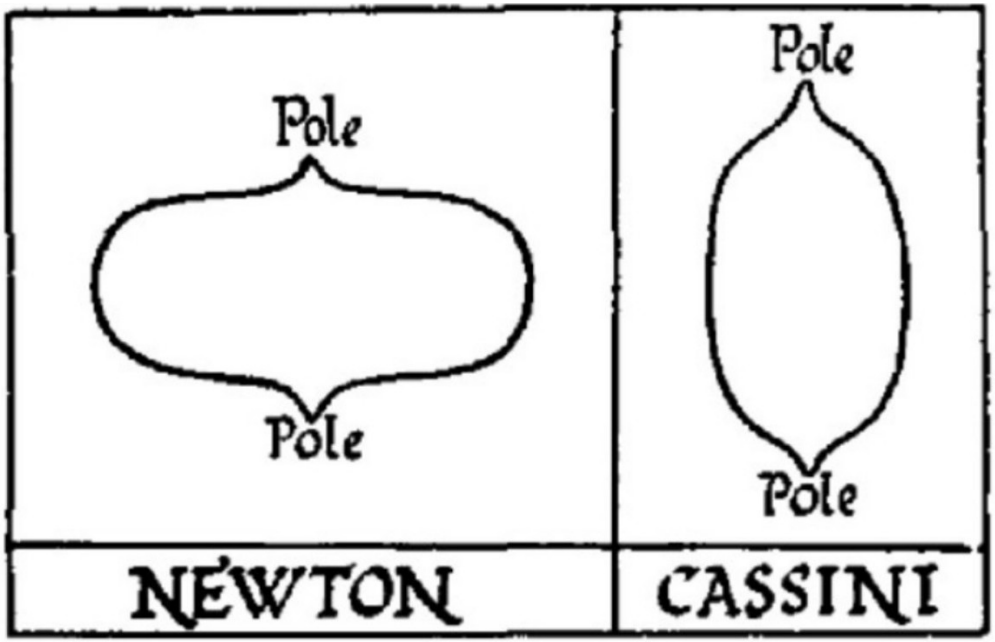
$$\mathcal{L}(\mathbf{x}, t, \boldsymbol{\varphi}, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}\|^2 - W(\mathbf{x}, t, \mathbf{F}) - \frac{1}{2}\rho(\mathbf{x})\zeta(\mathbf{x}, t).$$

Decomposition of the motion relative to an approximately rotating frame:

$$\varphi_i(\mathbf{x}, t) = \Phi_i(t) + R_{ij}(t)\overline{\varphi}_j(\mathbf{x}, t).$$

Equilibrium equations:

$$\rho\epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l x_m - \frac{\partial T^0_{ij}}{\partial x_j} - \rho\gamma^0_i = 0.$$



"Vouz avez confirmé dans les lieux pleins d'ennui
Ce que Newton connut sans sortir de chez lui."

Chandrasekhar (1969)

Solutions of the equilibrium equations

Notation from last lecture:

$$\rho\epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l x_m - \frac{\partial T^0_{ij}}{\partial x_j} - \rho\gamma^0_i = 0.$$

To simplify things we now write:

$$\rho\epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l x_m - \frac{\partial \sigma_{ij}}{\partial x_j} - \rho g_i = 0.$$

At equilibrium, referential and spatial quantities coincide.

The stress tensor is symmetric and so has six independent components.

The force balance holds in general materials, not just elastic ones.

Suppose the planet's shape, density and angular velocity are known.

We then have **three** equations for the **six** components of stress:

$$\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial \sigma_{ij}}{\partial x_j} - \rho g_i = 0.$$

This is an underdetermined system and so we **cannot expect unique solutions**.

More formally, there exist **non-trivial divergence-free symmetric second-order tensor fields** that satisfy the correct boundary conditions.

Compare with the existence of divergence-free vector fields.

Hydrostatic equilibrium

The (Cauchy) stress in a compressible Newtonian fluid:

$$\sigma_{ij} = -p\,\delta_{ij} + \eta\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{2}{3}\frac{\partial v_k}{\partial x_k}\delta_{ij}\right).$$

At equilibrium, the stress is **hydrostatic**:

$$\sigma_{ij} = -p\,\delta_{ij}.$$

The same result holds in more general viscous fluids.

Scaling estimate for deviatoric stress associated with convection:

$$\eta V / L \sim 10^6\,\mathrm{Pa}.$$

Pressures in the Earth's interior are of order $10^{10}\,\mathrm{Pa}$ – deviation from hydrostatic state in Earth-like planets represents only a **small perturbation**.

Given the hydrostatic approximation, we have:

$$\rho\epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l x_m + \frac{\partial p}{\partial x_i} - \rho g_i = 0.$$

This is now an **overdetermined** set of equations – three equations for one unknown.

Helpful to switch to a vector-calculus notation:

$$\nabla p + \rho\nabla\gamma = \mathbf{0}, \qquad \gamma = \phi + \psi.$$

Here we have introduced the **gravity potential**, this being the **sum of the gravitational and centrifugal potentials**.

Hydrostatic equilibrium equation:

$$\nabla p + \rho \nabla \gamma = \mathbf{0}.$$

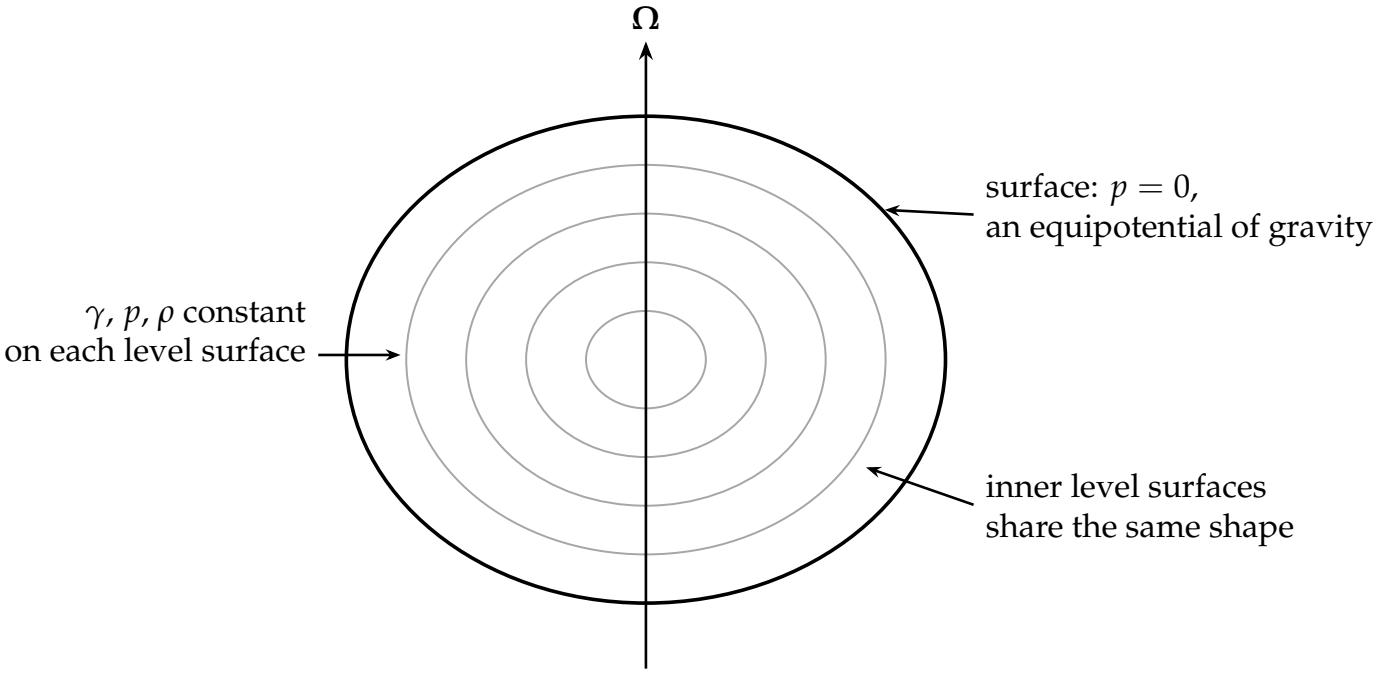
Gradients of pressure and gravity potential are parallel, and so a level surface of one field is a level surface of the other.

Ignoring the atmosphere, the pressure vanishes at the planet's surface, and hence this must be a level surface of the gravity potential.

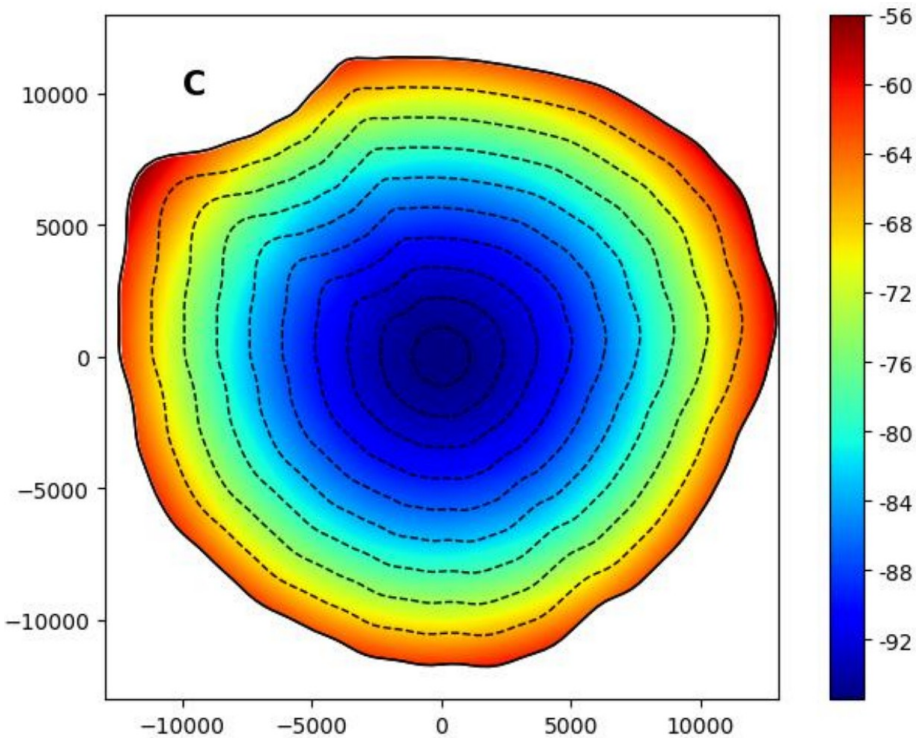
Taking the curl of the equation, we find:

$$\nabla \rho \times \nabla \gamma = \mathbf{0}.$$

This shows that level surfaces of density are also coincident with the other two fields.



In hydrostatic equilibrium the level surfaces of p, ρ and γ all coincide, and the surface is an equipotential of gravity.



Equatorial slice through the gravitational potential for Phobos calculated assuming a constant interior density.

The potential is not constant on the moon's surface, and hence this cannot be a hydrostatic equilibrium state.

In general, the hydrostatic equilibrium condition imposes very strong conditions on a planet's shape, density, and rotation.

Myhill *et al.* (2025)

In the non-rotating case, suppose the planet is spherically symmetric.

Equilibrium equation reduces to:

$$\nabla p + \rho \frac{\mathrm{d}\phi}{\mathrm{d}r} \hat{\mathbf{r}} = \mathbf{0}.$$

There is just one non-trivial equation here, and so we can find an equilibrium solution:

$$\frac{\mathrm{d}p}{\mathrm{d}r} + \rho \frac{\mathrm{d}\phi}{\mathrm{d}r} = 0, \qquad p(b) = 0.$$

It can be shown that this is the **only** solution in the non-rotating case.

By definition, a planet is **slowly rotating** if its centrifugal potential is small relative to its gravitational potential:

$$\frac{\text{centrifugal potential}}{\text{gravitational potential}} \sim \frac{3\Omega^2}{8\pi G\bar{\rho}}.$$

For the Earth, we find:

$$\frac{\text{centrifugal potential}}{\text{gravitational potential}} \sim 2 \times 10^{-3}.$$

In such cases, we can apply first-order perturbation theory.

Higher-order theories exist and are sometimes necessary.

Expand about the non-rotating solution, with s a perturbation parameter:

$$\begin{aligned}\rho &= \rho_0 + s\rho_1 + \cdots, \\ p &= p_0 + s\,p_1 + \cdots, \\ \phi &= \phi_0 + s\,\phi_1 + \cdots, \\ \psi &= s\,\psi_1.\end{aligned}$$

Similarly, the free-surface radius is expanded as:

$$r(\theta,\varphi) = b + s\,h_1(\theta,\varphi) + \cdots.$$

Topography on internal boundaries is expanded similarly, but we neglect this detail for simplicity in what follows.

First-order perturbation of the equilibrium equation:

$$\nabla p_1 + \rho_0 \nabla \gamma_1 + \rho_1 g_0 \hat{\mathbf{r}} = \mathbf{0},$$

with $\gamma_1 = \phi_1 + \psi_1$ and $g_0 = \mathrm{d}\phi_0/\mathrm{d}r$.

Exact boundary condition on the perturbed planetary surface:

$$p\left[b + s\, h_1(\theta, \varphi) + \cdots, \theta, \varphi\right] = 0.$$

Expanded to first order, and using $\mathrm{d}p_0/\mathrm{d}r = -\rho_0 g_0$, this reduces to:

$$p_1 + \frac{\mathrm{d}p_0}{\mathrm{d}r} h_1 = 0,$$

$$p_1 - \rho_0 g_0 h_1 = 0.$$

First-order perturbation of the equilibrium equation:

$$\nabla p_1 + \rho_0 \nabla \gamma_1 + \rho_1 g_0 \hat{\mathbf{r}} = \mathbf{0}.$$

Take the cross product with the radial unit vector:

$$\hat{\mathbf{r}} \times \nabla p_1 + \hat{\mathbf{r}} \times \rho_0 \nabla \gamma_1 = \mathbf{0},$$

$$\hat{\mathbf{r}} \times \nabla (p_1 + \rho_0 \gamma_1) = \mathbf{0}.$$

Taking the perturbations to average to zero over spherical surfaces, we conclude that:

$$p_1 = -\rho_0 \gamma_1.$$

First-order perturbation of the equilibrium equation:

$$\nabla p_1 + \rho_0 \nabla \gamma_1 + \rho_1 g_0 \hat{\mathbf{r}} = \mathbf{0}.$$

Taking the curl of this equation, we arrive at:

$$\hat{\mathbf{r}} \times \nabla \left(\frac{\mathrm{d}\rho_0}{\mathrm{d}r} \gamma_1 - g_0 \rho_1 \right) = \mathbf{0}.$$

And hence, we see that:

$$\rho_1 = \frac{1}{g_0} \frac{\mathrm{d}\rho_0}{\mathrm{d}r} \gamma_1.$$

We have now shown that the following identities hold for the first-order perturbations:

$$p_1 = -\rho_0 \gamma_1, \quad h_1 = -\frac{\gamma_1}{g_0}, \quad \rho_1 = \frac{1}{g_0}\frac{\mathrm{d}\rho_0}{\mathrm{d}r}\gamma_1.$$

All other terms can be expressed through the gravity potential perturbation.

Linearisation of Poisson's equation is trivial:

$$\nabla^2\phi_1 = \begin{cases} 4\pi G\rho_1 & r < b, \\ 0 & r > b. \end{cases}$$

Linearisation of the continuity conditions requires more work:

$$[\phi_1]^+_- = 0, \quad [\hat{\mathbf{r}} \cdot \nabla\phi_1 + 4\pi G\rho_0 h_1]^+_- = 0.$$

Derivation given in an appendix to the notes (non-examinable).

Expressing other perturbations in terms of the gravity potential:

$$\nabla^2\phi_1 = \begin{cases} 4\pi G\dfrac{1}{g_0}\dfrac{\mathrm{d}\rho_0}{\mathrm{d}r}(\phi_1+\psi_1) & r<b, \\ 0 & r>b, \end{cases}$$

along with the continuity conditions:

$$[\phi_1]^+_- = 0, \qquad \left[\hat{\mathbf{r}}\cdot\nabla\phi_1 - 4\pi G\rho_0\frac{(\phi_1+\psi_1)}{g_0}\right]^+_- = 0.$$

Linear boundary value problem for the perturbed gravitational potential, with the centrifugal potential providing the “force terms”.

Theory applies also to equilibrium tides in gaseous planets.

The centrifugal potential is proportional to Y_{20}, and separation of variables shows that:

$$\phi_1 + \psi_1 = \sqrt{\frac{4\pi}{5}}\frac{2}{3} r g_0 \epsilon \, Y_{20},$$

which serves to define the **ellipticity**, $\epsilon(r)$. Figures of slowly rotating planets are oblate spheroids.

At the planet's mean surface (and at internal ones too) we have:

$$\epsilon(b) = \frac{\text{equatorial radius} - \text{polar radius}}{\text{mean radius}}.$$

Surface ellipticity of the Earth is around 1/300.

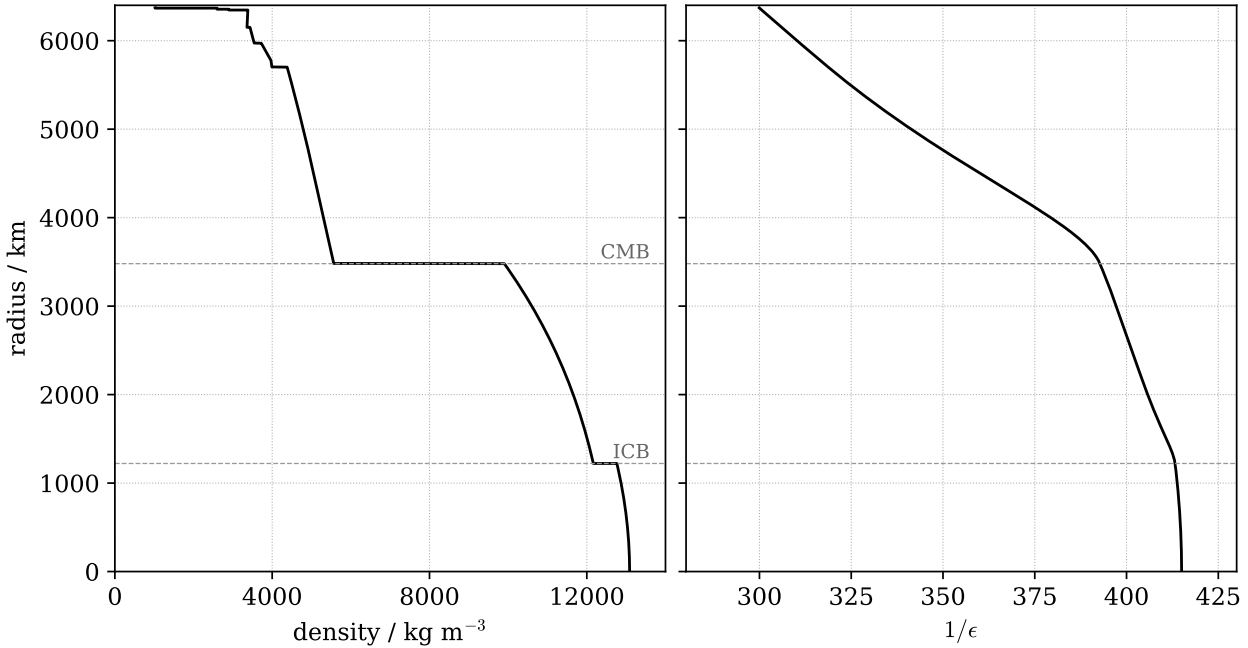
The ellipticity satisfies **Clairaut's equation** (Clairaut 1743):

$$\frac{\mathrm{d}^2\epsilon}{\mathrm{d}r^2} + 8\pi G\rho_0 g_0^{-1}\left(\frac{\mathrm{d}\epsilon}{\mathrm{d}r} + \frac{\epsilon}{r}\right) - \frac{6\epsilon}{r^2} = 0,$$

subject to the boundary conditions:

$$\left.\frac{\mathrm{d}\epsilon}{\mathrm{d}r}\right|_{r=0} = 0, \quad \left.\frac{\mathrm{d}\epsilon}{\mathrm{d}r}\right|_{r=b} = \frac{1}{b}\left[\frac{5\Omega^2 b^3}{2GM} - 2\epsilon(b)\right].$$

Solved numerically by integrating outwards from $r = 0$ and rescaling to meet the condition at $r = b$.



Left: radial density profile obtained from analysis of the Earth's free oscillations – final two lectures.

Right: the corresponding ellipticity (or flattening), plotted here as a reciprocal.

Hydrostatic surface value 1/299.9, observed 1/298.3: the difference reflects non-hydrostatic stresses in the mantle.

What is a planet?



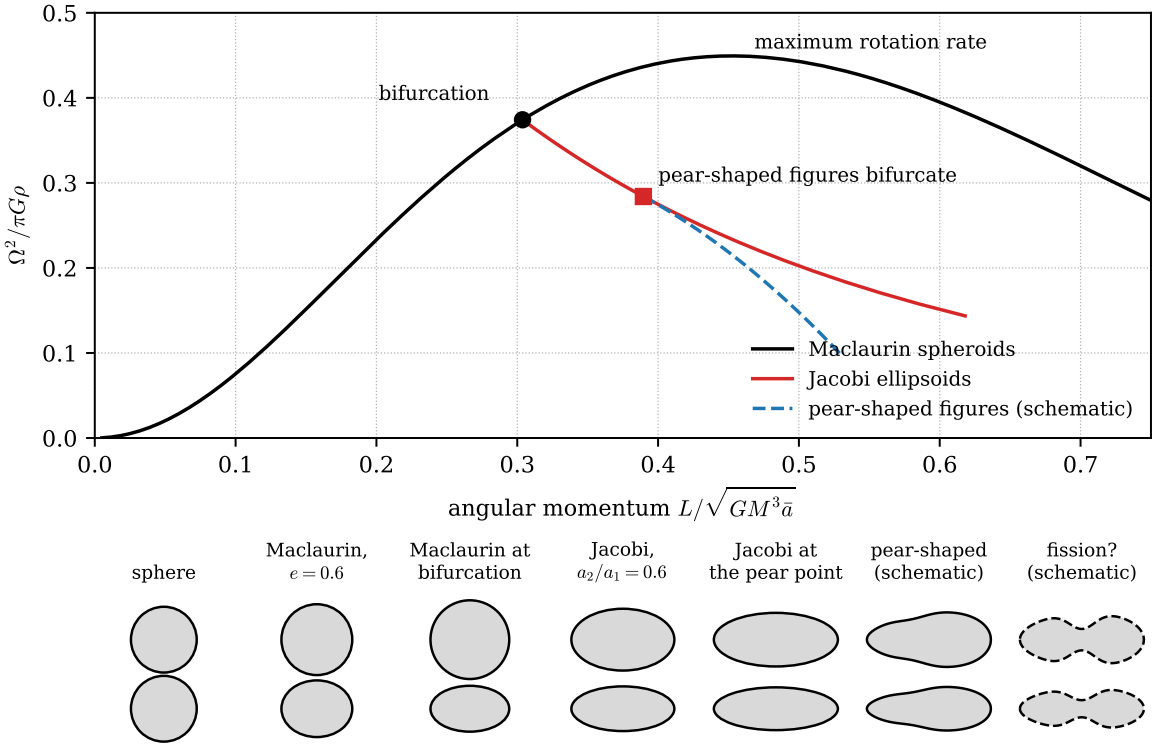
A planet



Not a planet

NASA

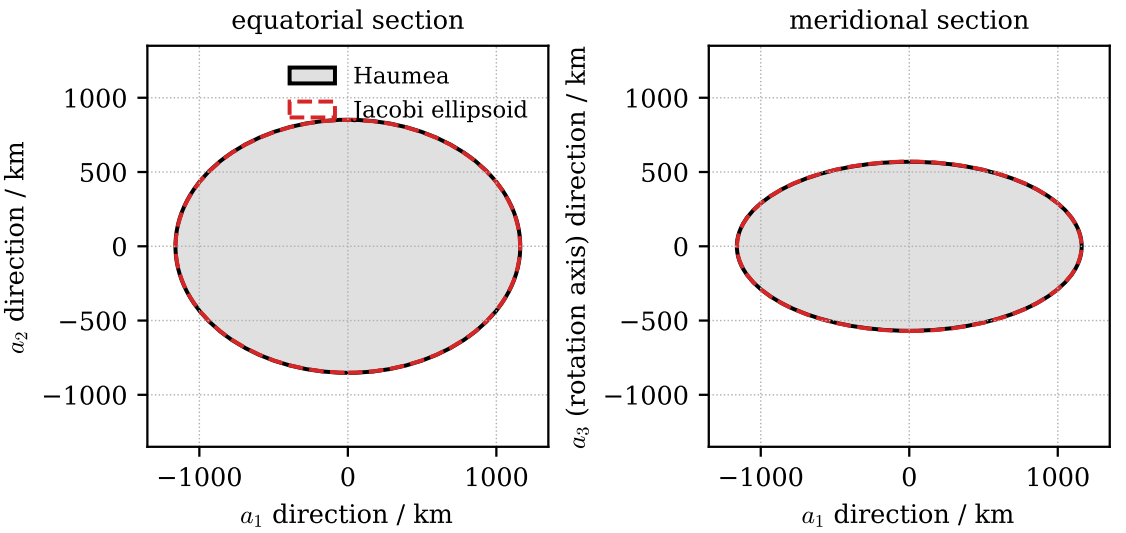
Rapidly rotating bodies (non-examinable)



Homogeneous fluid at any rotation rate: **Maclaurin spheroids** (1742) and **Jacobi ellipsoids** (1834).

The Jacobi sequence **bifurcates** from the Maclaurin sequence; pear-shaped figures branch in turn from the Jacobi sequence.

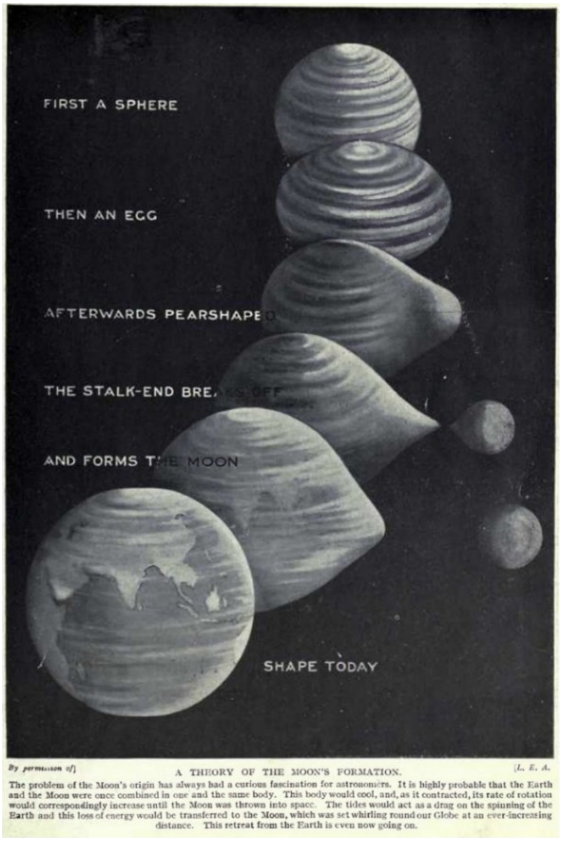




Equilibrium figures **do not have to be axisymmetric**.

Haumea is a dwarf planet with a rotation period of 3.9 hours and a triaxial figure, its axes within 1 % of a Jacobi ellipsoid (Ortiz *et al.* 2017).

It rotates too fast to be a homogeneous fluid, and so must have a denser core.



By permission of] A THEORY OF THE MOON'S FORMATION. [L. E. A.

The problem of the Moon's origin has always had a curious fascination for astronomers. It is highly probable that the Earth and the Moon were once combined in one and the same body. This body would cool, and, as it contracted, its rate of rotation would correspondingly increase until the Moon was thrown into space. The tides would act as a drag on the spinning of the Earth and this loss of energy would be transferred to the Moon, which was set whirling round our Globe at an ever-increasing distance. This retreat from the Earth is even now going on.

Fission hypothesis for the Moon's formation, due to George Darwin. Long dismissed in favour of a giant impact, but it does have a real physical basis, and it has recently attracted renewed interest.

What you need to know and be able to do

(i) Know the form of the equilibrium equations, and that they hold in a general planet and not just an elastic one.

(ii) That the equilibrium equations are underdetermined, and why.

(iii) What a hydrostatic equilibrium state is, and be able to derive the constraints imposed on the level surfaces of p, ρ, and γ.

(iv) Derive the hydrostatic equilibrium state in a non-rotating planet.

(v) Know in outline how solutions can be obtained within a slowly rotating planet. Here you will not need to reproduce the full derivation, but may be asked to perform related calculations with appropriate guidance.